

Novel interpretable Feature set extraction and classification for accurate atrial fibrillation detection from ECGs

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ARTICLE INFO

Keywords:

Atrial fibrillation
Electrocardiography (ECG)
Wavelet transform
Correlation
Machine learning
XGBoost

ABSTRACT

Objective: We present a novel method for detecting atrial fibrillation (AFib) by analyzing Lead II electrocardiograms (ECGs) using a unique set of features.

Methods: For this purpose, we used specific signal processing techniques, such as proper orthogonal decomposition, continuous wavelet transforms, discrete cosine transform, and standard cross-correlation, to extract 48 features from the ECGs. Thus, our approach aims to more effectively capture AFib signatures, such as beat-to-beat variability and fibrillatory waves, than traditional metrics. Moreover, our features were designed to be physiologically interpretable. Subsequently, we incorporated an XGBoost-based ECG classifier to train and evaluate it using the publicly available 'Training' dataset of the 2017 PhysioNet Challenge, which includes 'Normal,' 'AFib,' 'Other,' and 'Noisy' ECG categories.

Results: Our method achieved an accuracy of 96 % and an F1-score of 0.83 for 'AFib' detection and 80 % accuracy and 0.85 F1-score for 'Normal' ECGs. Finally, we compared our method's performance with the top-classifiers from the 2017 PhysioNet Challenge, namely ENCASE, Random Forest, and Cascaded Binary. As reported by Clifford et al., 2017, these three best performing models scored 0.80, 0.83, 0.82, in terms of F1-score for 'AFib' detection, respectively.

Conclusion: Despite using significantly fewer features than the competition's state-of-the-art ECG detection algorithms (48 vs. 150–622), our model achieved a comparable F1-score of 0.83 for successful 'AFib' detection. **Significance:** The interpretable features specifically designed for 'AFib' detection results in our method's adaptability in clinical settings for real-time arrhythmia detection and is even effective with short ECGs (<10 heartbeats).

1. INTRODUCTION

Atrial Fibrillation (AFib) is one of the most common cardiac arrhythmias, affecting nearly 2 % of the adult population worldwide [1]. AFib has been correlated with a 4-to-6-fold increased risk of ischemic stroke [2–5] and an increased risk of heart attack. Considered a 'silent killer,' one-third of all AFib cases are asymptomatic. AFib is characterized by erratic electrical activity causing irregular atrial contraction rates of up to 500 heartbeats per minute (bpm) [1]. AFib typically begins with short episodes that, if left untreated, get progressively longer, and can become permanent. Thus, early detection of AFib is critical to

prevent severe cardiac events that lead to increased morbidity and mortality. Typically, AFib is detected using electrocardiography (ECG) and subsequent clinical evaluation of the PQRST waves [6]. Key identifiers of AFib in an ECG include beat-to-beat variability, the absence of the P-wave, and the presence of fibrillatory waves [7].

Many methods have been proposed to detect AFib from ECG signals automatically [8–14]. Early studies have primarily focused on the morphological parameters of the ECG signal, including those analyzing ventricular response-specific factors (e.g., R–R interval) [15–17], as well as ones analyzing atrial activity (e.g., absence of P-wave and presence of fibrillatory f-waves) [18,19]. Since ECG signals are non-stationary and

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nonlinear, some studies have explored the spectral behaviors of AFib using time-frequency analysis, such as wavelet transforms [20]. However, these studies have failed to adequately combine the morphological aspects, such as the beat-to-beat variation, with the spectral behavior analysis. Hence, these studies utilized a few interpretable features but demonstrated limited classification accuracy. Currently, ECG classification studies focused on incorporating machine learning (ML) methods [21–29], especially convolutional neural networks (CNN) or hybrid methods like CNN-LSTM [22,25–33]. However, ML methods require a significant number (>150) of features to maintain high classification accuracy, resulting in the loss of interpretability (or physiological relevance) of the features [9,10,15,17,34]. Moreover, despite the ability of ML techniques to extract features and perform classification automatically, they suffer from data over-fitting and data bias, as well as insufficient training data sources, reducing accuracy and reliability [28,30,31,36]. Further, current ML algorithms require longer ECGs (>100 heartbeats) [37], limiting their clinical applicability. Overall, the landscape of ECG classification has evolved significantly with the advent of sophisticated Machine Learning (ML) architectures. The most recent techniques have increasingly adopted advanced methods such as Transfer Learning and Generative Adversarial Networks (GAN) [Weimann et al., 2021; Golany et al., 2024] [38,39]. These state-of-the-art approaches have shown promise in improving the accuracy and robustness of ECG classification. However, they come with considerable challenges. One major issue is the algorithmic complexity inherent in these deep learning-based models [Zahid et al., 2023; Xiao et al., 2022; Rahman et al., 2022; Wang et al., 2024] [40–43]. Such complexity necessitates extensive training on powerful Graphics Processing Units (GPUs), a resource seldom available in typical clinical environments, especially in rural areas. Moreover, while lightweight models have been proposed as alternatives to these resource-intensive techniques, they often rely heavily on machine-extracted deep features. These features, while useful, lack the physiological interpretability that carefully crafted, hand-engineered features provide [Zahid et al., 2023; Xiao et al., 2022] [40,41]. This reliance can limit the clinical utility of such models, as they do not offer the transparency required for clinicians to understand and trust the diagnostic outputs. Methods that combine multiple ML models into a single algorithmic pipeline can enhance robustness. While this approach can improve performance, it also adds layers of complexity, making the system harder to interpret and deploy in a clinical setting [Cai et al., 2020] [44]. Each added layer or combined model increases the computational burden and difficulty integrating with existing medical infrastructure. In contrast, our method offers a more streamlined solution. We utilize an XGBoost model, which is a lightweight ML approach that significantly reduces the need for high-level computational resources. XGBoost is known for its efficiency and performance, even on less powerful hardware, which makes it particularly well-suited for practical applications in clinical environments. This simplicity does not come at the expense of accuracy. Instead, our model leverages a carefully developed set of features that maintain physiological relevance, ensuring that the system is both interpretable and reliable. Thus, our approach meets the computational constraints typical of clinical settings and provides a robust and understandable tool for ECG classification, enhancing its practical applicability.

In this paper, we propose a method to classify ECGs by combining a novel cohort of interpretable features with an ML classifier, thereby addressing the current limitations of such methods. However, the primary objective of this study was to develop and validate a novel, interpretable, and efficient method for detecting atrial fibrillation (AFib) from electrocardiogram (ECG) signals. The method leverages a unique set of physiological features combined with an XGBoost-based classifier. The study utilized the publicly available dataset from the 2017 PhysioNet [45], which comprises > 8k single-lead ECG recordings. These recordings are categorized into four classes: Normal, AFib, Other, and Noisy, with each recording lasting between 0.5 and 1 min. The core components of the algorithm are chronologically described below.

- **ECG data pre-processing:** This was done using signal denoising and QRS complex detection: The discrete wavelet transform (DWT) was employed to denoise the ECG signals. This method effectively removes baseline wander, power line interference, and other artifacts while preserving the essential characteristics of the signal. Then, feature extraction was performed: The continuous wavelet transform (CWT) was used to detect the QRS complexes, which are crucial for segmenting the ECG signal into individual heartbeats.
- **Feature Extraction:** A total of 48 features were extracted from the ECG signals, grouped into four categories: 1. Correlation-based features capture beat-to-beat variability and include peak width ratios and normalized peak widths. 2. ECG Morphology features such as median beat length, PR interval, QT interval, and heart rate variability were included to provide insights into the structural characteristics of the ECG signals. 3. Discrete Cosine Transform (DCT)-Based Features were used to capture frequency-domain characteristics of the ECG signals. 4. Entropy-based feature measures were used to quantify the complexity and irregularity of the ECG signals.
- **Model Training and Evaluation:** An XGBoost classifier was chosen for its efficiency and performance in handling structured data. An exhaustive grid search was conducted to optimize the hyperparameters of the XGBoost model, including maximum depth, minimum child weight, gamma, alpha, lambda, the number of estimators, and learning rate. Resampling techniques were applied to address class imbalance. The Normal and Other classes were down-sampled, and k-fold cross-validation was used to ensure robust performance evaluation. The model was evaluated using accuracy, F1-score, sensitivity, specificity, and precision to comprehensively assess its performance across different classes.

The rationale for selecting these approaches is summarized here: The discrete wavelet transform (DWT) preserves important signal characteristics while eliminating noise, such as baseline wander and power line interference, common in ECG signals. The continuous wavelet transform (CWT) provides excellent time-frequency localization, making it ideal for accurately detecting QRS complexes essential for heartbeat segmentation. The correlation-based features capture beat-to-beat variability, which is a hallmark of AFib. ECG Morphology Features provide detailed information on the structural aspects of heartbeats, which is crucial for distinguishing between normal and abnormal rhythms. DCT-based features capture frequency-domain characteristics, offering a different perspective from time-domain features. Entropy-based features quantify the complexity and irregularity of the ECG signals, highlighting the random nature of AFib. XGBoost is a robust, efficient, and interpretable machine learning algorithm that performs well with structured data. It also handles imbalanced datasets effectively and does not require extensive computational resources, making it suitable for practical clinical applications. By employing these methods, we create a system that is accurate, efficient, interpretable, and practical for real-time clinical applications. The proposed method integrates the strengths of DWT and CWT for signal processing with the robust yet lightweight XGBoost classifier to offer a dependable tool for the early and precise detection of AFib.

2. Method description

ECG classification methods aim to detect AFib and other arrhythmias by identifying differentiating characteristics, or features, between the abnormal and normal signals and using ML to classify signals based on these features [45–47]. However, noise corruption of ECG signals, irregular heartbeats, and imperceptibility of abnormal patterns reduce the accuracy of automated algorithms [8,34,47]. Thus, the extracted features' quality and efficacy affect ML models' success [40].

We introduce a novel feature set that uses continuous wavelet transforms (CWT) with cross-correlation analysis to identify and quantify differences in the underlying ECG signal structure. Unlike Fast

Fourier Transform (FFT), choosing CWT provides temporal and spectral information from the signal. Lastly, an optimized XGBoost ECG classifier was trained by supervised learning based on these features. A general algorithm flow chart is shown in Fig. 1, and the processing steps are described in the following paragraphs.

2.1. ECG pre-processing

2.1.1. 1) signal denoising

The discrete wavelet transform (DWT) was used to denoise the ECG signals, following the procedure detailed in Martis et al. [48,49]. The DWT decomposes the signal into wavelet basis functions resulting in a multi-level filter bank which is then used to reconstruct a frequency-filtered signal. Since the baseline resides in the low-frequency band of 0–2 Hz, and frequencies above 45 Hz contain no useful information [35], the reconstruction retained only the 2–45 Hz range. To denoise the ECG signal, we used a Daubechies mother wavelet (db6) due to its shape similarities to an ECG signal and reconstructed the 3rd to 6th-level coefficients.

2.1.2. 2) QRS-complex detection & filtering

To analyze an ECG signal, we first need to identify the QRS complex. The QRS complex represents the electrical activity during ventricle contraction [50]. Since the QRS complex exhibits localized high-frequency components, the continuous wavelet transform (CWT) is excellent for extracting this signal portion [50]. To detect the QRS complex, the Hilbert transform of the zero-mean denoised ECG signal is computed to obtain the analytic signal, and the CWT of this analytical signal is obtained using the db6 mother wavelet. The Hilbert transform leads to a smooth scalogram (Fig. 2b). The inverse CWT (ICWT) of the normalized wavelet coefficients is then computed using the Morse mother wavelet. Mixing the mother wavelet used for each operation amplifies the R-peaks while minimizing noise, making the R-peaks easier to identify (Fig. 2c). Identified R-peaks are filtered based on the peak height, peak prominence, and the half peak width. Any R-peaks occurring on an underlying residual wave are removed to avoid obscuring the peak location. The Morse wavelet provides excellent time-frequency localization, which is crucial for accurately reconstructing ECG signals. This property allows for precise identification and reconstruction of the QRS complex, P waves, and T waves in the ECG, which are essential for detecting atrial fibrillation (AFib). While other mother wavelets, such as Daubechies or Morlet, are also commonly used in ECG signal processing, the Morse wavelet offers a unique combination of properties that make it particularly suitable for our needs.

The locations of the Q and S waves are identified using the R-peak information and the CWT-filtered version of the ECG signal. The CWT minimizes extraneous peaks caused by ancillary frequencies, and the Q and S-peaks become the most prominent minima (Fig. 1).

The identified QRS complexes are then filtered. A QRS complex is considered too noisy if the R-peak height falls outside the 95 % confidence interval of all R-peak measurements or if the QRS length deviates more than 45 % from the median. Additionally, QRS complexes are removed based on HR variability (with a tolerance of 20 % from the median) and the variability of all ECG heartbeats (with a tolerance of 20 % from the mean). The latter is calculated as the average width of the cross-correlation between QRS complexes of a given beat and all other heartbeats. Further filtering is done by computing the signal-to-noise ratio (SNR) of the filtered signal and removing heartbeats with an estimated noise level of the denoised ECG greater than 33 %.

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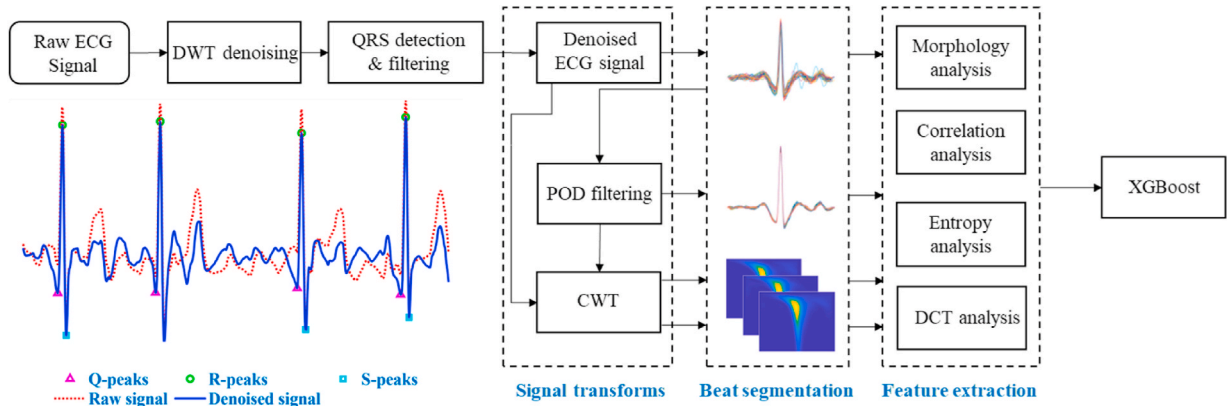


Fig. 1. Schematic of the proposed ECG classification method.

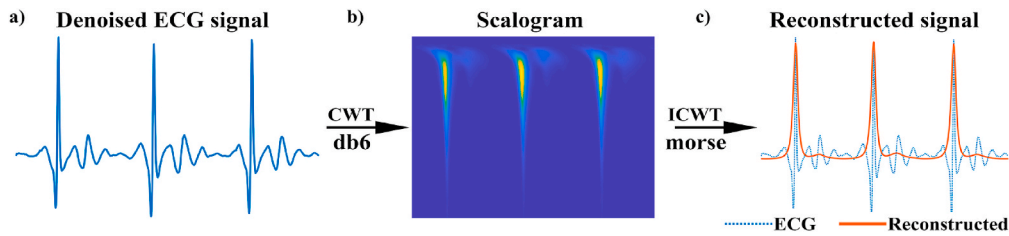


Fig. 2. Extraction of R-peaks with help of CWT and ICWT calculations. a) A three-beat sample segment of the denoised ECG signal. b) The scalogram of the denoised ECG signal using the db6 mother wavelet. c) The reconstructed signal from the scalogram using the morse mother wavelet.

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2.2.3. 3) heartbeats segmentation

Each heartbeat's start and end points were defined using the mid-points between adjacent R-peak locations. The ECG signal was subsequently segmented into heartbeats, and the median beat length was computed. Any ECG location with a non-QRS complex signal value greater than 70 % of the average R-peak height was considered 'noisy' and excluded from further analysis. The R-peak in an ECG signal represents the most prominent feature, typically having the highest amplitude. By setting the threshold at 70 % of the average R-peak height, we aim to distinguish true signal components from noise. Non-QRS components exceeding this threshold are likely artifacts or noise rather than physiological features, ensuring that noise does not interfere with the analysis. So, the 70 % threshold for excluding non-QRS complex signal values as 'noisy' during heartbeat segmentation is justified by its empirical validation, effectiveness in distinguishing noise from signal, alignment with noise characteristics in ECG recordings, and consistency with existing literature. This threshold ensures the preservation of signal integrity while effectively mitigating the impact of noise on the analysis, contributing to the overall robustness and accuracy of the proposed method. To ensure accuracy, we remove heartbeats that are affected by noisy ECG signal locations or deviate too much from the median beat

length. We then calculate a metric that compares each beat to the median beat in terms of point-by-point difference and R peak height and location. Any remaining erroneous heartbeats are identified and removed based on a plateau of this metric. Ultimately, we retain between 85 % and 95 % of all heartbeats.

The segmented heartbeats were separated into two half-signals, namely the signal before and after the QRS complex to identify the P and T-waves. Since P and T-waves differ in duration, amplitude, and noise surrounding the peak, different criteria were applied to find each peak [51,52]. The P-wave peak was defined as the maximum positive peak of the first half-beat signal. To find the T-wave, the peaks of the latter half-beat were calculated. Peaks were eliminated based on the threshold of 25 % and 33 % of all peaks' maximum height and prominence, respectively. The T-peak was the farthest of the remaining peaks from the QRS complex.

2.3. ECG representations

The segmentation process comprises one representation (Representation 1) of the ECG beat signals. To create more representations of the segmented ECG heartbeat signals, we use two signal processing techniques, as illustrated in Fig. 1. First, we apply the CWT using the db6 mother wavelet to the denoised ECG signal. Then, we segment the resulting scalogram using the previously computed beat information to generate Representation 2.

Proper orthogonal decomposition (POD) is applied to the denoised and segmented ECG beat signals. POD decomposes each beat into eigenmodes sorted by decreasing energy [53]. A reduced-order model of the signals can be produced by reconstructing a subset of the eigenmodes [54]. Fig. 3 b shows the segmented beat signals before and after the POD filtering operation for a normal and AFib ECG signal. Using the 'POD filtered' beat signals, we reconstructed these to a single (non-segmented) ECG signal, applied the CWT, and subsequently segmented the resulting scalogram into heartbeats (Representation 4).

The various representations of the ECG beat signal enable a more robust and detailed evaluation of the ECG beat signal. For example, Representation 1 is primarily effective for quantifying beat-to-beat variability of the R-peaks, while Representation 3 suppresses the R-peak variability such that it can be used to evaluate fibrillatory wave variability. In addition to these four representations, we apply the POD operation to the QRS-only and the QRS-removed denoised ECG signals. Further, the scalogram of the denoised ECG signal is segmented in intervals of two heartbeats to analyze the variability of irregular heartbeats.

2.4. Feature extraction

We extract 48 features from the ECG signal and beat representations. The features can be grouped in four categories based on the signal processing methods used to extract the given feature: correlation-based features (C), ECG.

morphology features (M), discrete cosine transform-based features (D), and entropy-based features (E). For robustness, each feature category maintains redundant analysis that focus on the AFib hallmark

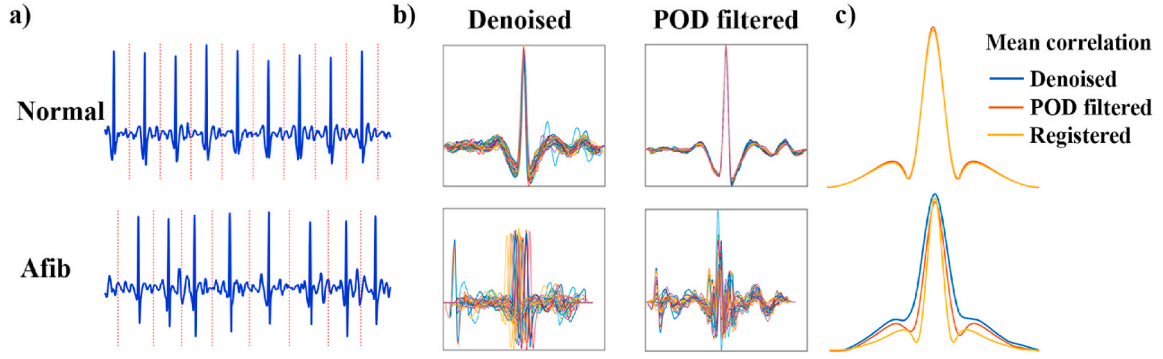


Fig. 3. a) Sample denoised ECG signal segments for Normal and AFib. b) Segmented, denoised ECG heartbeats (Representation 1) and POD filtered signal heartbeats (Representation 3). c) 1-D projection in the time domain of the correlation plane of the denoised (Representation 2), POD-filtered (Representation 4), and registered scalograms for a Normal and AFib case.

characteristics of beat-to-beat variability and the presence of fibrillatory waves.

Table 1 provides the group and description of all features.

2.4.1. 1) correlation-based features

Correlation analysis provides a means to quantify irregularities across heartbeats. We apply standard cross-correlation (SCC) to the two segmented ECG scalogram representations (Representations 2 and 4) and the two-beat segmented scalogram stack. The correlation was applied between each beat and the mean beat for that signal representation. Using the SCC results, we register the Representation 2 heartbeats. Since irregular AFib heartbeats are expected in the early disease stages, SCC was also computed between the temporally adjacent heartbeats of the registered scalograms to investigate irregularities among the adjacent heartbeats. Fig. 3c illustrates the 1-D projection in the time domain of the mean correlation plane for the denoised (Representation 2), POD-filtered (Representation 4), and registered scalogram heartbeats for a Normal and AFib case. Fig. 3b and c highlight that the irregularity of the AFib heartbeats results in different correlation plane widths across the signal representations. The primary correlation-based features used for signal classification include the peak width of the resulting correlation planes and peak width ratios comparing the width across ECG beat representations. Correlation plane widths were computed using 1-D projections in the time and frequency domains. A total of 22 correlation-based features are calculated.

2.4.2. 2) ECG morphology features

The identified P-wave, QRS complex, and T-wave were used to compute ECG morphology features, including the features currently most often used in clinical diagnostics. These features include median beat length, normalized PR interval, normalized QT interval, and heart rate variability. A total of 13 morphology-based features are calculated.

2.4.3. 3) DCT-based features

The DCT represents a signal as a sum of cosine functions with various amplitudes and frequencies [55]. The main advantage of DCT over other mathematical transforms is that DCT offers high energy compaction, resulting in fewer coefficients to describe the signal [55]. Thus, the DCT provides an excellent tool for classification based on energy compaction since AFib ECG signals contain significantly more variance than normal ECGs, as observed in Fig. 3b. Hence, the 2-D DCT was applied to the denoised and POD-filtered beat signal stacks (Representations 1 and 3). The 1-D mean DCT coefficients of all heartbeats were computed, and an inverse model ($y = \frac{a}{x^b}$) was used to this mean signal. The DCT analysis was done for the denoised ECG and QRS-removed, denoised ECG signal. A total of 10-DCT-based features are extracted.

2.4.4. 4) entropy-based features

The spectral entropy of the CWT and DCT of the QRS-removed ECG signal is calculated to investigate the information in the P and T-waves. The spectral entropy was computed according to the Shannon entropy equation,

$$H = -\sum_i x_i \log_2 x_i \quad (1)$$

where x_i represents the probability distribution of the power spectrum of CWT and DCT coefficients. A total of three entropy-based features are computed.

2.5. XGBoost classifier

2.5.1. 1) Feature normalization

Once feature extraction was completed, a supervised machine-learning classifier model was incorporated in the algorithm pipeline using *eXtreme Gradient Boosting* (XGBoost). First, the features in the training dataset were normalized to remove the variability in the nominal values by scaling their numeric values to a common scale. This step also helps reduce data redundancy and increase convergence rate. Since the training dataset's cumulative distribution function (CDF) does not follow a Gaussian pattern, the normalization is performed using the z-score method (standardization). This method scales the features according to the:

$$Z = \frac{x - \mu}{\sigma} \quad (\text{Eq. 2})$$

where x is the original feature vector and μ , σ are the features' mean and standard deviation, respectively. Moreover, using z-score standardization prevents outliers from affecting the data classes.

2.5.2. Hyper-parameter optimization

The optimum performance of the XGBoost model was ensured by hyper-parameter tuning using an exhaustive grid search since the decision trees of XGBoost have minimized loss functions. Thus, we obtained the best-fitted model among all possible combinations for the feature set. We tuned 7 booster parameters, including maximum decision tree depth (Max^m Depth), minimum child weight (Min^m Child Weight), number of decision trees (No. of Estimators), model learning rate, L-1 minimizer loss (α), L-2 minimizer loss (λ), and minimum loss for tree pruning (γ). Table 2 provides the ranges, step sizes, and final values used in the parameter tuning. All other general and learning task parameters of the XGBoost model were set to their initial default values.

2.5.3. 3) removing class imbalance

Resampling techniques were applied to address the class imbalance in the ECG dataset. Specifically, the Normal and Other ECG classes

Table 1

Extracted features categorized to four groups based on the type of analysis namely, correlation analysis (C), morphology analysis (M), Discrete cosine transform analysis (D), and entropy analysis (E).

No.	Feature Description	Group	No.	Feature Description	Group
1	Peak width ratio between denoised and registered scalogram correlation projected to time domain	C	25	Mean error between DCT and the inverse model with denoised signal beats	D
2	Normalized peak width of denoised scalogram correlation projected to time domain	C	26	Mean error between DCT and the inverse model with POD filtered signal beats	D
3	Peak ratio between registered and POD filtered scalogram correlation projected to time domain	C	27	Numerator (a) of DCT inverse model with denoised signal beats	D
4	Peak ratio between POD filtered and registered scalogram correlation projected to frequency domain	C	28	Exponent (b) of DCT inverse model with QRS-removed denoised signal beats	D
5	Cumulative deviation between all correlations	C	29	Exponent (b) of DCT inverse model with QRS-removed denoised signal beats after POD filtering	D
6	Normalized peak width of POD filtered scalogram correlation projected to time domain	C	30	Mean error between DCT and the inverse model with QRS-removed denoised signal beats	D
7	Normalized P-wave width based on correlation with median P-wave	C	31	Mean error between DCT and the inverse model with QRS-removed denoised signal beats after POD filtering	D
8	Correlation area of P wave of POD filtered beats with median P wave	C	32	Numerator (a) of DCT inverse model with QRS-removed denoised signal beats	D
9	Normalized T-wave width based on correlation with median T-wave	C	33	Spectral entropy of the CWT with QRS-removed denoised signal	E
10	Correlation area of T wave of POD filtered beats with median T wave	C	34	Spectral entropy of the DCT with QRS-removed denoised signal	E
11	Normalized peak width of two-beat scalogram correlation projected to time domain	C	35	Spectral entropy of the DCT with QRS-removed denoised signal after POD filtering	E
12	Area under two-beat scalogram correlation projected to time domain	C	36	Median beat length	M
13	Asymmetry metric of two-beat scalogram correlation projected to time domain	C	37	Heart rate variability	M
14	Predominant peaks ratio of two-beat scalogram correlation projected to time domain	C	38	Normalized QRS interval with average beat length	M
15	Peak width of denoised beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	39	Normalized PR interval	M

Table 1 (continued)

No.	Feature Description	Group	No.	Feature Description	Group
16	Peak width of POD filtered beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	40	Normalized PR interval using POD filtered P-wave	M
17	Asymmetry metric of denoised beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	41	Standard deviation of P-peak locations using POD filtered P-wave	M
18	Predominant peaks ratio of denoised beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	42	Normalized QT interval	M
19	Asymmetry metric of POD filtered beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	43	Normalized QT interval using POD filtered T-wave	M
20	Predominant peaks ratio of POD filtered beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	44	Standard deviation of T-peak locations using POD filtered T-wave	M
21	Normalized mean secondary peaks prominence of POD filtered beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	45	Normalized standard deviation of R-peaks using median	M
22	Normalized peaks width difference of POD filtered beat stack autocorrelation projected to time domain with QRS-removed denoised signal	C	46	Normalized total absolute deviation of R-peaks using median	M
23	Exponent (b) of DCT inverse model with denoised signal beats	D	47	Median R peak ratio of QRS-only denoised signal beats after POD filtering	M
24	Exponent (b) of DCT inverse model with POD filtered signal beats	D	48	Noise percentage of QRS-removed denoised signal	M

Table 2

Settings of XGBoost model's hyper-parameter tuning procedure.

Booster Parameters	Value Range	Step Size	Tuned Values
Max ^m Depth	3~10	1.0	10
Min ^m Child Weight	0.1~1.0	0.1	1
Gamma (γ)	0.0~0.05	0.1	0
Alpha (α)	10^{-5} ~ 10^2	Log 10	0
Lambda (λ)	0.0~1.0	0.1	1
No. of Estimators ($n_{estimators}$)	50~350	50	100
Learning Rate, Eta (ξ)	10^{-4} ~ 10^{-1}	Log 10	10^{-4}

underwent down-sampling before cross-validation. The removed samples were then utilized in additional iterations of k-fold cross-validation, and the results were obtained by averaging the outcomes of all iterations.

2.6. Testing data and evaluation metrics

2.6.1. 1) ECG recordings used for testing

We tested our ECG classifier using the ‘2017 PhysioNet/CinC (Computing in Cardiology) Challenge’ dataset (<https://physionet.org/content/challenge-2017/>) [45]. It consists of 8582 ECG recordings ranging from 30 s as the median length to a maximum of 60 s. All the recordings, sampled at 300 Hz, were generated using AliveCor, an FDA (U.S. Food and Drug Administration)-approved and medical-grade personal ECG device. The recordings include 5050 ECGs classified as ‘Normal,’ 738 ECGs as ‘AFib,’ 2456 as ‘Other,’ and 284 as ‘Noisy.’ This public dataset is considered the ‘training’ set from the PhysioNet Challenge. A ‘testing’ set with additional recordings was used for the challenge. However, because the ‘testing’ set is neither public nor shareable, we split the ECG recordings in the PhysioNet ‘training’ dataset into separate training and testing sets to assess the performance of our proposed method.

2.6.2. 2) k-fold cross-validation

For enhanced generalization, k-fold cross-validation was used for creating k number of partitions in the original dataset. Later, each unique subset was divided into 80-20 train-test splits. Thus, the test sets were hidden from the train sets for k-folds of iteration. The fold size (here, k = 5) was selected to avoid model overfitting and computational complexity. The evaluation metrics from different folds were averaged to get the final output of the classifier.

2.6.3. 3) evaluation metrics

Four evaluation metrics were used to measure the classifier’s performance: accuracy, F1 score, sensitivity, specificity, and precision. Accuracy was computed according to the following:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (3)$$

where TP, FP, TN, and FN stand for True-Positive, False-Positive, True-Negative, and False-Negative, respectively. Accuracy demonstrates the ratio of the correctly labeled ECG classes to the whole pool of ECG data.

F1-score (or F-measure) was computed according to the:

$$\text{F1 score} = \frac{2 * (PPV * TPR)}{(PPV + TPR)} \quad (4)$$

Here, PPV, Positive Predictive Value (also called “precision”), represents the likelihood that ECG data belong to a specific class when the model labels it as such.

On the other hand, TPR (True Positive Rate) or sensitivity (also termed hit-rate, or “recall”), indicates the probability of labeling a given class correctly. Sensitivity is calculated according to.

The F1-score, which is the harmonic mean of precision and recall, combines these measures to assess the model’s performance.

Specificity is the probability of true negative, given that the classifier correctly classifies it. Specificity or true negative rate (TNR) is calculated as follows:

$$\text{Specificity (TNR)} = \frac{TN}{TN + FP} \quad (6)$$

3. RESULTS

3.1. A. algorithm performance summary

Table 3 summarizes the performance of our method for the Normal,

Table 3

Evaluation metrics for our algorithm across all four categories: ‘Normal’, ‘AFib’, and ‘Other’.

Metrics	Overall	Normal	AFib	Other
Accuracy (%)	84.9 ± 0.04	80.4 ± 0.04	95.8 ± 0.02	78.6 ± 0.05
F1-score	0.75 ± 0.09	0.85 ± 0.04	0.83 ± 0.12	0.56 ± 0.12
Sensitivity	0.63 ± 0.05	0.72 ± 0.05	0.71 ± 0.03	0.47 ± 0.07
Specificity	0.72 ± 0.03	0.83 ± 0.02	0.81 ± 0.01	0.53 ± 0.04

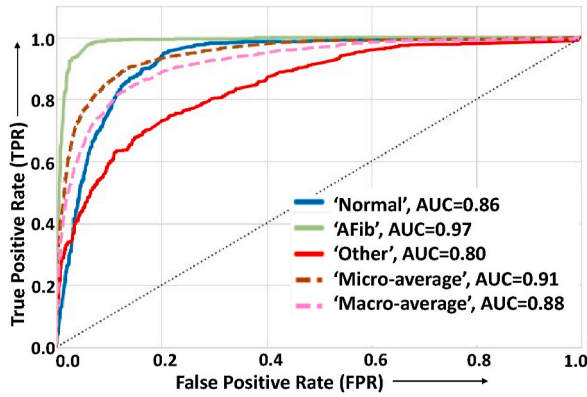
AFib, and ‘Other’ classifications (we omitted the Noisy category for not being scored in the 2017 PhysioNet Challenge). Our method achieved an accuracy of ~96 % and an F1-score of 0.83 for AFib detection. It identified Normal ECGs with an accuracy of 80 % and an F1-score of 0.85 and maintained accuracy for the ‘Other’ category at ~79 % and F1-score of 0.56. The sensitivity and specificity were 0.47 and 0.53 for the ‘Other’ cases. This suggests that our method was primarily accurate for the Normal and AFib cases. Since the Other case for this dataset contains any other arrhythmia besides AFib, the Other ECGs maintain high variability across samples. Hence, our features, focused on AFib characteristics, struggled to identify other arrhythmias. However, the significant difference in the performance for Normal and AFib classes can be attributed to several factors related to the nature of the ECG signals and the extracted feature set. First, AFib is characterized by erratic electrical activity, leading to irregular atrial contractions. This manifests in the ECG as beat-to-beat variability and the presence of fibrillatory waves, distinct and more easily identifiable features. In contrast, normal ECG signals exhibit regular and consistent patterns, making subtle variations in these patterns more challenging to detect. Second, the feature set we extracted is tailored to capture the hallmark characteristics of AFib, such as beat-to-beat variability and the presence of fibrillatory waves. These features are highly discriminative for AFib detection. However, the variability is much lower for normal ECG detection, and the features may not be as distinct. This could reduce accuracy when distinguishing normal ECGs from other types, especially those classified under the ‘Other’ category. Also, referring to Table 4, it is observed that in most cases, normal ECG detection accuracy is higher than AFib detection accuracy (in this case, we had an imbalanced dataset: 5050 Normal (majority class) and 738 AFib (minority class) cases). This can be partially explained by the class distribution in the training dataset. The dataset contains a higher number of normal ECGs, providing the algorithm with more examples to learn from. Despite this, the features extracted for normal ECGs might overlap with those of other arrhythmias, leading to occasional misclassifications and a slight decrease in accuracy. Besides, the algorithm’s focus on AFib-specific features, while beneficial for detecting AFib, might not generalize as well to normal ECGs. Features like entropy and correlation-based metrics are designed to highlight irregularities, which are less prevalent in normal ECGs. As a result, the algorithm may perform exceptionally well in identifying AFib but less so in consistently recognizing normal rhythms, particularly when subtle deviations are present. To address these performance differences, future work could incorporate additional features that better capture the nuances of normal ECG patterns. Ensuring a more balanced representation of all classes (in this case, we had 5050 Normal and 738 AFib cases) to mitigate the effects of class imbalance. Exploring hybrid models that can leverage both AFib-specific and general ECG features to improve overall detection accuracy. Therefore, while the algorithm demonstrates high accuracy for AFib detection, the complexity and subtlety of normal ECG signals present unique challenges. By refining the feature set and addressing class imbalances, we might achieve more consistent performance across all ECG categories.

Other than the four metrics presented in Table 3, such as accuracy, F1-score, sensitivity, and specificity (introduced in Table 3), the ROC curves (probability curves) and corresponding AUC scores (measurement of degree of separability) are reported as the 5th metric. This information is represented in Fig. 4 for the three diagnostic ECG

Table 4

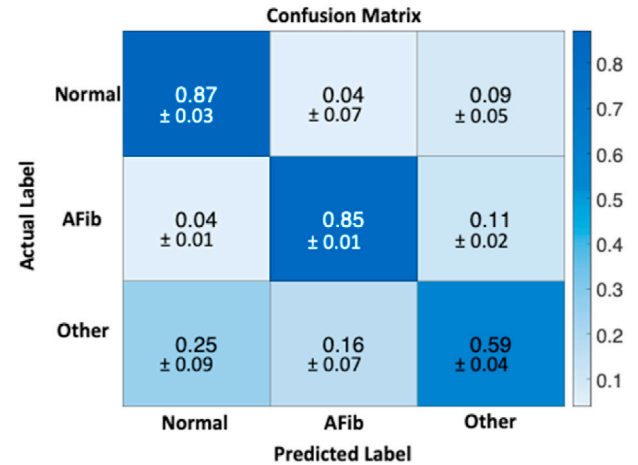
Comparison between the proposed model and the other state-of-the-art models on PhysioNet 2017 Challenge Dataset in terms of their F1-scores.

Classifier Model	Training Score of ECG Data Type				Test Score of ECG Data Type			
	Overall	Normal	AFib	Other	Overall	Normal	AFib	Other
ENCASE [11] (Hong et al., 2017)	0.83	0.92	0.84	0.74	0.82	0.91	0.80	0.76
Random Forest [12] (Zabihi et al., 2017)	0.82	0.90	0.79	0.75	0.82	0.90	0.83	0.73
Cascaded Binary [13] (Datta et al., 2017)	0.82	0.90	0.79	0.77	0.83	0.92	0.82	0.75
XGBoost Classifier (Proposed model)	0.83	0.89	0.97	0.64	0.75	0.85	0.83	0.56

**Fig. 4.** ROC curves and AUC scores for Normal, AFib, and Other diagnostic categories, and corresponding micro- and macro-average ROC curves.

categories. Classifiers that give curves closer to the top-left corner indicate a better performance. So, at the baseline or threshold, the ECG classifier is supposed to give points along the left diagonal where $TPR = FPR$. The further away the curve goes from the 45° diagonal of the ROC space, the more robust the test. On the other hand, $AUC = 0.5$ suggests no discrimination between the distribution, and the higher the AUC, the better the model is at predicting. From Fig. 4, it is evident that the ROC curve for AFib is in the top-left corner and furthest away from the threshold line, so a 97 % AUC score suggests reliable AFib detection. The micro and macro averages were also included to show the effects of diagnostic category imbalances on the classification results. The less bias our model holds towards the majority categories, Normal and Other, against the minority class, AFib, the closer the macro and micro averages are. There is only a 3 % discrepancy between the micro and macro average ROC-AUC scores for our case. Hence, this demonstrates that the proposed classifier model can create a clear distinction between the ECG categories despite the imbalance in the dataset.

Fig. 5 shows the confusion matrix, tabulating the predicted label provided by our method vs. the true label: it assesses the rate of TP, TN, FP, and FN for each class as well as the Type I error (FP rate) and Type II error (FN rate). Among true AFib ECGs, any signals misclassified by our method were about three times more likely to be classified as Other than Normal cases. Our method maintained a Type-I error of 20 % and a Type-II error of 15 % for AFib. For true Normal ECGs, our method maintained Type-I and Type-II errors of 13 % and 29 %, respectively. The high Type II error for true Normal ECGs was almost wholly contributed by 'Other' cases misdiagnosed as Normal. For true 'Other' cases, our classifier had a Type-I error of 20 % and a Type-II error of 41 %. In total, the confusion matrix demonstrates the high accuracy of our method for identifying true Normal and AFib cases. However, our classifier often misclassified 'Other' cases as Normal ~25 % of the time and AFib ~16 % of the time. Again, this suggests that our features were well-suited for Normal and AFib detection but less for 'Other' arrhythmias.

**Fig. 5.** Confusion matrix for proposed method.

$$\text{Sensitivity (TPR)} = \frac{TP}{TP + FN} \quad (5)$$

3.2. B. extracted Feature analysis

In Fig. 6, we investigate the efficacy of our novel feature set. In Fig. 6a–f, the probability density functions (PDF), and the common area overlap between the PDFs of the Normal, AFib, and 'Other' categories are shown for six features. The kernel density estimation method was employed to create all PDFs using the normal distribution kernel. Datapoints greater than 10^5 and less than -10^5 were omitted from the PDF calculation to remove outliers. Common area overlap represents the distinguishability between PDFs. A lower area overlap implies better distinguishability and thus highlights feature with better classification ability. Fig. 6a–c (top row) shows three of the novel correlation and DCT analysis features, and Fig. 6d–f (middle row) illustrates morphology features commonly used in medical diagnosis. The best features for differentiating between Normal and AFib were feature 2 (28 % overlap, correlation-based feature), feature 37 (34 % overlap, morphology-based feature), and feature 28 (53 % overlap, DCT-based feature). Among the traditional features, the 'Normalized QT interval' (feature 42) and 'median beat length' (feature 36) had larger overlap areas of 66 % and 78 %, respectively. Between AFib and Other, feature 37 (morphology feature), feature 1 (correlation feature), and feature 2 (correlation feature) were the best features with overlapping areas of 46 %, 47 %, and 53 %, respectively. The 'Normalized QT interval' (feature 42) and 'median beat length' (feature 36) again had a higher overlap of 87 % and 76 %, respectively. Furthermore, the overlap area for 'Normal' and 'Other' for all features was higher than that involving the AFib category. This highlights that the algorithm has higher differentiability of AFib vs. any other diagnosis.

Fig. 6g–i provides the common overlap area between the Normal, AFib, and 'Other' PDFs based on the feature category (morphology, correlation, DCT, and entropy) and XGBoost importance. Two features were excluded due to binary distribution, and their PDFs could not be

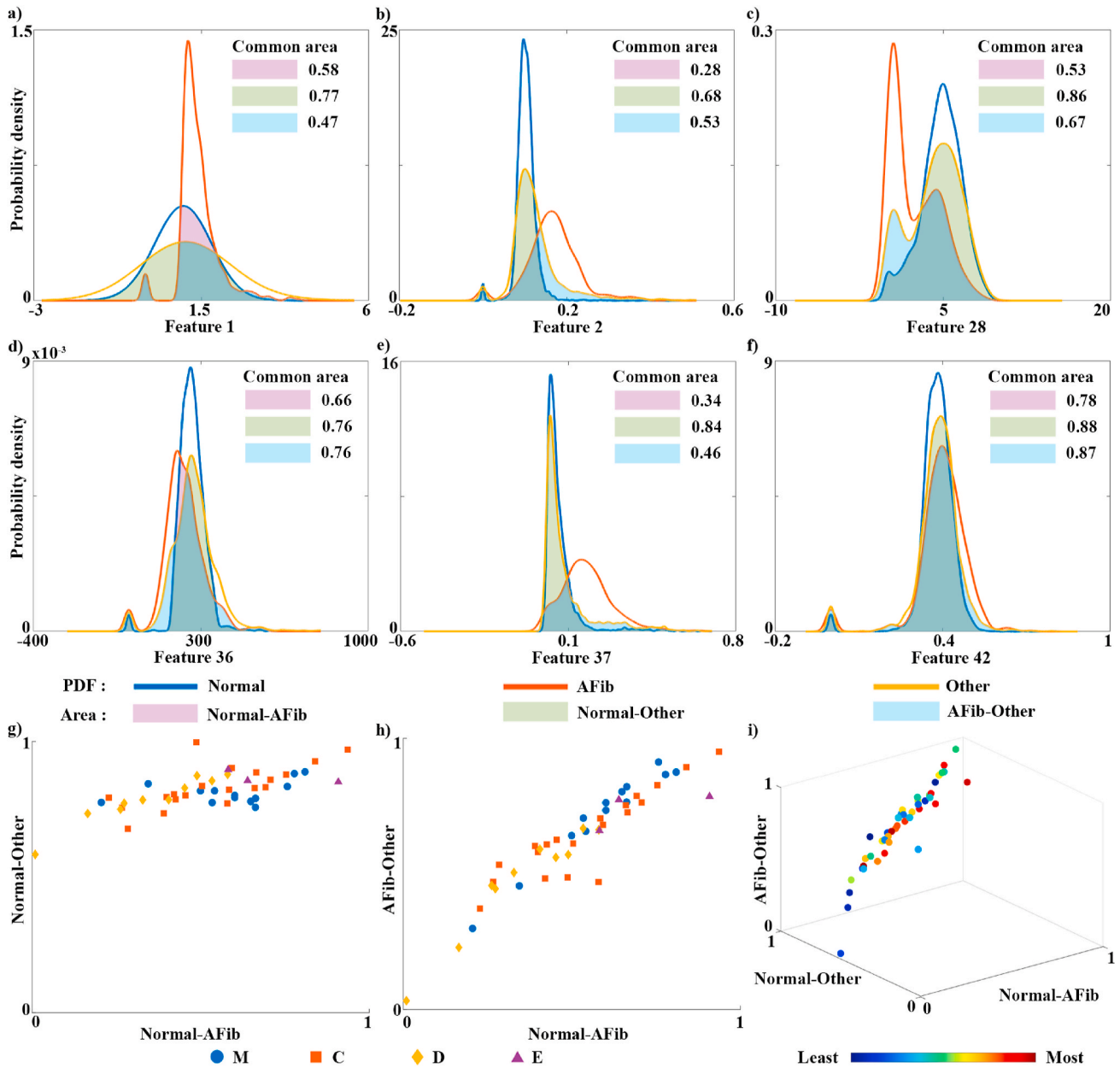


Fig. 6. The probability distribution functions (PDF) and the common overlap area under PDFs for features. a) feature 1 (peak width ratio between denoised and registered scalogram correlation projected to time domain). b) feature 2 (normalized peak width of denoised scalogram correlation projected to time domain). c) feature 28 (exponent (b) of DCT inverse model with QRS-removed denoised signal heartbeats). d) feature 36 (median beat length). e) feature 37 (Heart rate variability). f) feature 42 (Normalized QT interval). g) Overlap area under Normal-AFib and Normal-Other PDFs for features. h) Overlap area under Normal-AFib and AFib-Other PDFs for features. i) 3D scatter plot of overlap area under Normal-AFib, Normal-Other, and AFib-Other PDFs with XGBoost feature importance.

computed. Between Normal and AFib, the average overlap among the correlation, DCT, and morphology features were 54 %, 35 %, and 59 %, respectively. The average overlap area between AFib and Other was 65 % for correlation features, 47 % for DCT features, and 72 % for morphology features. Finally, an average overlap of 83 %, 79 %, and 81 % for correlation, DCT, and morphology features were obtained between 'Normal' and 'Other'. These results highlight that for AFib, the novel correlation and especially DCT features maintained better distinguishability than traditional morphology features, implying the effectiveness of the features. Furthermore, a minimum overlap area difference of approximately 9 % was observed between our novel, proposed features, and the standard features for Normal vs. AFib and AFib vs. Other. However, this difference was reduced to less than 1 % in the case of Normal and Other. Hence, although our novel feature set excelled at

AFib identification it underperformed to differentiate between Normal and Other cases.

Lastly, redundant analysis plays a crucial role in ensuring the robustness and reliability of our model. By incorporating multiple features that may overlap or capture similar information, we enhance the model's ability to generalize and perform well across varied and potentially noisy datasets. Redundancy helps stabilize the model by providing multiple pathways for capturing essential information, ensuring critical data is not lost even in the presence of noise. This approach improves generalizability by making the model more robust to data variations and reduces overreliance on single features, maintaining reliability in clinical settings with variable ECG signal quality. Furthermore, including redundant features supports a comprehensive analysis of the ECG signals, leveraging insights from correlation-based,

morphology-based, DCT-based, and entropy-based features to create a holistic and robust model.

3.3. C. comparison to other methods

We next compare the performance of our method with the highest performing, state-of-the-art classifiers from the 2017 PhysioNet Challenge. Specifically, we compare against the ENCASE [11] (Hong et al., 2017), Random Forest [12] (Zabihi et al., 2017), and Cascaded Binary [13] (Datta et al., 2017). Scores for these classifiers are reported in Clifford et al., 2017. For ENCASE (Hong et al., 2017), the F1-scores excluded the deep or uninterpretable features. We note a limitation of this analysis shown in Table 4: the scores for the three comparison classifiers are using the PhysioNet Challenge “Test Set”; however, because this set is neither public nor shareable, we cannot test our method using the same dataset and instead need to utilize the 80-20 split from the PhysioNet Challenge “Training Set”.

Among the three comparison models, the ‘ENCASE’ model uses 622 features, the ‘Random Forest’ model employs 491 features and down-selected 150 features, and the ‘Cascaded Binary’ model utilizes 150+ features. Hence, our model, at 48 features, uses the least. Despite the significantly smaller number of features, our model matched the best-performing algorithm for identifying AFib (F1-score of 0.83). However, the other models performed slightly better for Normal cases and ‘Other’ cases.

3.4. D. classification accuracy vs. ECG signal length

Because AFib is a progressive disease, automated detection algorithms must be accurate for early diagnosis, even for short ECG signals (<10 beat signals). Thus, Fig. 7 shows the accuracy and F1-score of our method as a function of the number of heartbeats within the ECG signal. As expected, a larger number of heartbeats (availability of more information) increases the accuracy of our classifier. It is observed that our model accuracy plateaus at a signal length of about 70 heartbeats, where additional heartbeats in the signal result in minimal increases in accuracy. Fig. 7 highlights that our method is accurate even for short beat lengths. Specifically, the reduction of ECG signal length from over 100 heartbeats to 10–20 heartbeats only resulted in a 10–15 % reduction in accuracy and F1-score.

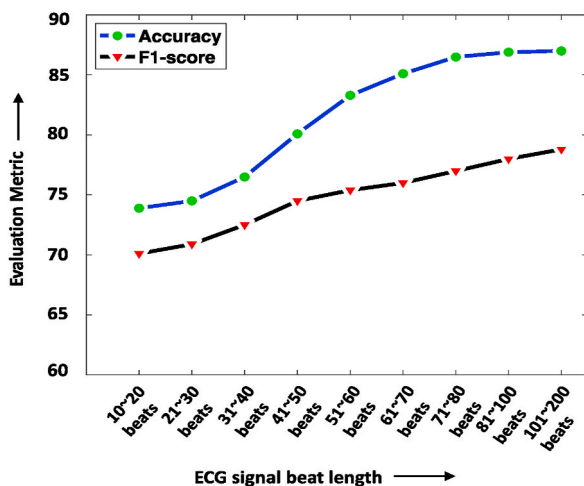


Fig. 7. Plot of evaluation metric values from ECG classifier output vs. ECG signal beat length from different ECG beat-groups: overall Accuracy and F1-score of the proposed classifier improves with increasing number of beats in ECGs.

4. DISCUSSION

In this study, we introduced an automated ECG classifier that utilized a novel, interpretable feature set combined with an XGBoost-based ML model. The features were designed based on the hallmark characteristics of AFib (e.g., irregular, and variable heartbeats and the presence of fibrillatory waves). Moreover, rather than using the typical clinical features such as R–R interval to assess these hallmark ECG characteristics, we incorporated signal processing tools for assessing the ECGs. We demonstrated that this approach vastly improved the efficacy and robustness of the feature set for identifying AFib in ECG signals. As a result, our method maintained AFib classification accuracy and F1 scores on par with the top-performing algorithms from the 2017 PhysioNet AFib Challenge, despite utilizing significantly fewer features. Moreover, the proposed method for detecting AFib from ECGs leverages a novel set of interpretable features and an XGBoost-based classifier, offering significant clinical benefits: (i) Early and Accurate Detection: Early detection of AFib is crucial for preventing serious complications such as stroke, heart failure, and other cardiovascular diseases. Our method achieves a high accuracy of ~96 % in detecting AFib, enabling timely intervention and improved patient outcomes through prompt treatment and management. (ii) Clinical Transparency, Trust, and Decision-Making: Our approach’s interpretable feature set ensures that the features are physiologically relevant and easily understandable by clinicians. This transparency facilitates better clinical decision-making and fosters trust in the algorithm’s predictions. (iii) Real-Time Detection: Using XGBoost, a lightweight ML model, ensures that our method is computationally efficient and does not require extensive resources. This makes it feasible for real-time applications in clinical settings, including integration with wearable devices for continuous monitoring. (iv) Robustness with Short ECG Segments Allows Early Diagnosis: Detecting AFib in short ECG segments is valuable for early diagnosis and monitoring during transient arrhythmic episodes. Our method demonstrates robustness with short ECG signals (as few as 10 heartbeats), making it suitable for early detection of AFib episodes and potentially reducing the need for prolonged monitoring sessions. (v) Accessibility in Resource-Limited Settings: The reduced computational cost and smaller model size (3.2 MB) make our approach accessible to a wide range of healthcare facilities, including those with limited technological infrastructure. (vi) Integration with Remote Monitoring and Wearable Devices: Our algorithm’s efficiency and accuracy make it well-suited for integration with remote monitoring systems and wearable ECG devices. This can enable continuous monitoring of patients with known AFib or those at high risk, improving ongoing care and reducing the need for frequent hospital visits. (vii) Personalized Care: The detailed and interpretable features extracted by our method can provide personalized insights into a patient’s cardiac health. This allows for more tailored treatment plans and monitoring strategies, aligning with the growing trend of personalized care in healthcare.

A major contribution of our method is this notion of interpretable features. More specifically, this refers to features that maintain physiological relevance and interpretability. Current ML-based ECG arrhythmia classifiers are based on uninterpretable or non-physiological features [29–44]. Thus, ML classifiers provide a binary output. As a result, clinicians cannot validate ML-based diagnoses bedside, which remains a main factor preventing ML classifiers from gaining clinical traction. Our method’s features were explicitly designed to provide physiological insight. For example, the correlation width and the DCT parameter represent beat-to-beat irregularity. The correlation peak width of the QRS-removed denoised ECG feature quantifies the presence of fibrillatory waves in the signal. Herein, we established for AFib that incorporating a few interpretable features, rather than many non-physiological features, did not result in a loss of classification accuracy. Moreover, the analysis demonstrated the distinguishability of specific features, highlighting that our approach can provide clinicians with information that can be interpreted instead of a “black-box”-type

diagnosis.

Another key advantage of the reduced number of features is that it reduces the overall computational cost and results in a smaller model size. Our hyperparameter-tuned and cross-validated XGBoost model size is only 3.2 MB, significantly reducing the processing time and making it a faster detection method. Specifically, using a standard CPU-based computer (Processor: 2.3 GHz Quad-Core Intel Core i7), it takes 50~55 s to train the about 8500 signal dataset. Typically, DNN (Deep-layered Neural Network) based classifier models have larger model sizes (as large as 1 GB) and rely on longer processing time [32]. Hence, this reduction in computational cost enhances our method's real-time and clinical utility.

Further, as demonstrated in Fig. 6, our method is accurate even for short ECG signals containing just 10–20 heartbeats. This implies that our method could identify AFib even in episodes that only last for as few as 10 heartbeats before normal sinus rhythm is restored. This capability can enable real-time arrhythmia detection. Our method for detecting AFib is designed to supplement and coincide with the clinician's assessment of cardiac arrhythmia progression, considering the patient's health history and any coexisting conditions.

The core contribution of this paper lies in the novel feature detection approach employed for accurate AFib detection from ECG signals. Out of the 48 features reported, certain features are particularly novel and contribute significantly to the classification performance. 1. Correlation-Based Features: Peak Width Ratio (F6): This feature measures the ratio of peak widths between different representations of the ECG signal, capturing the beat-to-beat variability characteristic of AFib. Normalized Peak Width (F20): This feature provides a normalized measure of peak widths, enhancing the detection of irregular heartbeats associated with AFib. 2. Discrete Cosine Transform (DCT)-Based Features: DCT Exponential (F32): The exponent in the inverse DCT model applied to the denoised signal beats captures the frequency-domain characteristics crucial for distinguishing AFib from normal rhythms. Mean Error (F43): This feature calculates the mean error between the DCT and its inverse model, highlighting the more pronounced discrepancies in AFib cases. 3. Entropy-Based Features: Entropy of Transform (F18): Entropy measures the complexity of the ECG signal, with higher values indicating more irregularity. This feature is particularly useful for identifying the chaotic nature of AFib signals. 4. Morphology-Based Features: Median Beat Length (F9): This feature measures the average duration of heartbeats, helping to distinguish the prolonged and irregular beats typical of AFib. PR Interval (F11): The normalized PR interval provides insights into the conduction times within the heart, which can be disrupted in AFib patients.

Contribution of Features to Classification: The feature importance plot (Fig. 8) highlights the relative contribution of each feature to the classification model. The weights of these features are derived from the F-score, which indicates their significance in the model. The following features stand out as the most influential: F6 (Correlation I C): This feature has the highest F-score (1539), indicating its critical role in capturing the variability and irregularity of heartbeats that signify AFib. F20 (Normalized I C): With an F-score of 1508, this feature effectively normalizes peak widths, making it a robust indicator of beat-to-beat variability. F32 (Exponential D): This feature's high F-score (1470) shows its importance in capturing frequency-domain characteristics that distinguish AFib. F18 (Entropy of T E): The entropy measure, with an F-score of 1384, underscores its value in identifying the complexity and irregularity of the ECG signals. F9 (Median Beat M) and F11 (PR Interval M): These morphology-based features, with F-scores of 1290 and 1260 respectively, are crucial for detecting the structural changes in ECG patterns associated with AFib [56].

Best-Performing Features: The best-performing features, as indicated by their F-scores, are predominantly those that capture the unique characteristics of AFib, such as beat-to-beat variability, frequency-domain anomalies, and signal complexity: Correlation-Based Features (F6, F20): These features are highly effective in detecting the irregular

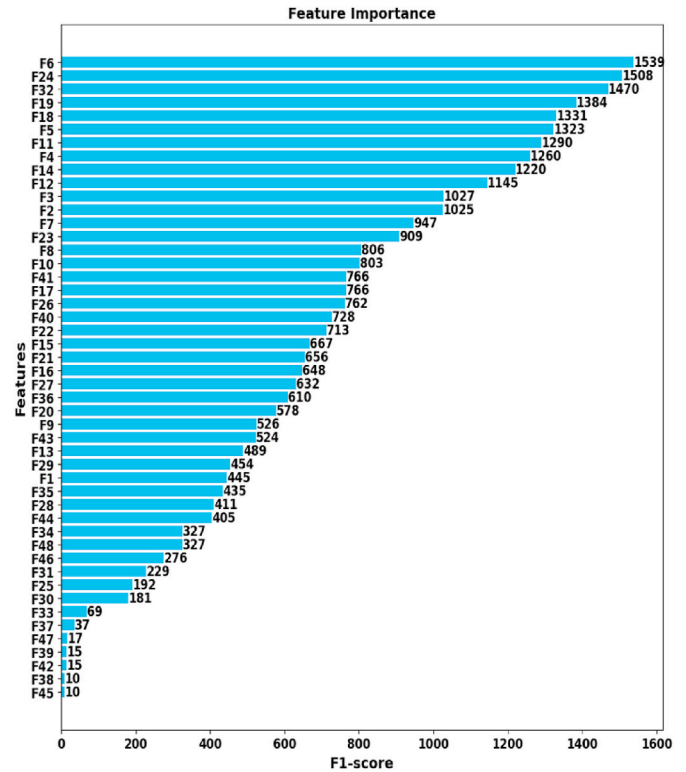


Fig. 8. Features used and their corresponding weights.

patterns of AFib. DCT-Based Features (F32, F43): These capture subtle frequency-domain changes that are not apparent in the time domain. Entropy-Based Feature (F18): This feature highlights the chaotic nature of AFib signals. Morphology-Based Features (F9, F11): These provide essential insights into the structural abnormalities in the ECG. These novel features proposed in this work significantly enhance the detection accuracy of AFib by capturing its unique characteristics. The high F-scores of these features underscore their importance in the classification model, making them key contributors to the algorithm's overall performance. By focusing on these features, our approach provides a robust, interpretable, and clinically relevant method for AFib detection from ECG signals [57].

Limitations and Future Work: Some limitations of our study include:

1. **Dataset Bias:** The dataset used in this study, the 2017 PhysioNet Challenge dataset, may not be fully representative of the broader population. This dataset might have inherent biases, such as demographic skewness (age, gender, ethnicity) and clinical variability (different types and stages of cardiac conditions). Such biases could affect the generalizability of our model. The algorithm's performance might be lower when applied to datasets with different demographic or clinical characteristics.
2. **Noise Handling:** Although discrete wavelet transform (DWT) and continuous wavelet transform (CWT) were employed for noise reduction and QRS detection, some noise artifacts might persist. The threshold for excluding noisy segments (>70 % of the average R-peak height) might not capture all instances of noise, leading to potential misclassifications. Residual noise can affect the accuracy of feature extraction and, consequently, the performance of the classification model.
3. **Feature Selection:** While we extracted a comprehensive set of 48 features, there might be other relevant features not included in our analysis. The selected features, although informative, might not capture all aspects of the ECG signal necessary for optimal classification. Excluding potentially significant features could limit the model's accuracy and robustness.
4. **Model Complexity and Interpretability:** Although XGBoost is relatively interpretable compared to other machine learning models, it still involves a degree of complexity that might hinder full

transparency. Clinicians might find it challenging to fully understand the model's decision-making process. Limited interpretability could affect the trust and acceptance of the model in clinical practice. 5. Real-Time Implementation: The proposed method, while efficient, has not been extensively tested in real-time clinical settings. Factors such as computational delays, integration with existing clinical workflows, and system user-friendliness were not fully explored.

To mitigate these limitations, possible future directions of our study include: 1. Addressing Dataset Bias: Future research could aim to validate the model on more diverse and larger datasets. Collecting and incorporating data from different populations, healthcare settings, and demographic groups can help enhance the model's generalizability. This ensures that the model performs well across varied patient groups and clinical conditions. 2. Enhanced Noise Handling: Future research might aim to develop more sophisticated noise reduction techniques and adaptive thresholds for identifying and excluding noisy segments. Techniques such as adaptive filtering and machine learning-based noise detection could be explored. This is to improve the accuracy of ECG signal preprocessing and feature extraction, leading to better classification performance. 3. Expanded Feature Set: Future research might go towards exploring and incorporating additional features that capture more aspects of the ECG signal. Advanced signal processing techniques and domain knowledge from cardiology can help identify new, relevant features. This enhances the model's ability to distinguish between different ECG patterns and improve overall accuracy. 4. Improving Model Interpretability: Investigating methods to increase the interpretability of the XGBoost model, such as feature importance visualization, decision path analysis, and incorporating explainable AI techniques, could be another potential future scope. This ensures that clinicians can understand and trust the model's predictions, facilitating its adoption in clinical practice. 5. Real-Time System Testing: This study has the potential to implement and test the model in real-time clinical settings, focusing on integration with existing systems, user interface design, and response times.

5. CONCLUSION

The proposed method for detecting atrial fibrillation (AFib) from electrocardiograms (ECGs) demonstrates high accuracy and efficiency, leveraging a novel set of interpretable physiological features combined with an XGBoost-based classifier. Achieving approximately 96 % accuracy for AFib detection and around 80 % for normal ECG detection, this method addresses the limitations of deep learning-based models by offering a lightweight solution that does not require extensive computational resources, making it suitable for real-time clinical applications. The novelty of our approach lies in its carefully crafted feature set, designed to capture the unique characteristics of AFib and ensure model interpretability and robustness. The practical implications are significant, with the method being ideal for integration into wearable devices and remote monitoring systems, facilitating early diagnosis and timely intervention. Future research should focus on validating the model with diverse datasets, enhancing noise reduction techniques, expanding the feature set, improving model interpretability, and testing in real-time clinical settings. The method's accuracy, robustness, and clinical utility can be further enhanced by addressing these areas, ultimately contributing to better cardiac care and patient outcomes.

Disclosure information

This work was funded by the National Institutes of Health (NIH), National Institute of Child Health & Human Development (NICHD) (Grant Number: 1R21HD109490). There are no other financial disclosures to report.

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Ruhi Sharmin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Melissa C. Brindise:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Jibin Joy Kolliyl:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Brett A. Meyers:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Jiacheng Zhang:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology. **Pavlos P. Vlachos:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] N. Henzel, J. Wróbel, K. Horoba, *Atrial Fibrillation Episodes Detection Based on Classification of Heart Rate Derived Features*, 2017.
- [2] P.A. Wolf, R.D. Abbott, W.B. Kannel, Atrial fibrillation as an independent risk factor for stroke: the Framingham Study, *Stroke* 22 (8) (1991) 983–988, <https://doi.org/10.1161/01.STR.22.8.983>.
- [3] J.S. Healey, et al., Subclinical atrial fibrillation and the risk of stroke, *N. Engl. J. Med.* 366 (2) (Jan. 2012) 120–129, <https://doi.org/10.1056/NEJM0A1105575>.
- [4] S.H. Hohnloser, et al., Incidence of stroke in paroxysmal versus sustained atrial fibrillation in patients taking oral anticoagulation or combined antiplatelet therapy: an ACTIVE W Substudy, *J. Am. Coll. Cardiol.* 50 (22) (Nov. 2007) 2156–2161, <https://doi.org/10.1016/J.JACC.2007.07.076>.
- [5] A.J. Camm, G. Corbucci, L. Padeletti, Usefulness of continuous electrocardiographic monitoring for atrial fibrillation, *Am. J. Cardiol.* 110 (2) (Jul. 2012) 270–276, <https://doi.org/10.1016/J.AMJCARD.2012.03.021>.
- [6] S.S. Chowdhury, M.S. Hasan, R. Sharmin, Robust heart rate estimation from PPG signals with Intense motion artifacts using Cascade of adaptive filter and Recurrent neural network, in: *IEEE Region 10 Annual International Conference, Proceedings/TENCON*, 2019–October, Oct. 2019, pp. 1952–1957, <https://doi.org/10.1109/TENCON.2019.8929692>.
- [7] X. Du, et al., A novel method for real-time atrial fibrillation detection in electrocardiograms using multiple parameters, *Ann. Noninvasive Electrocardiol.* 19 (3) (May 2014) 217–225, <https://doi.org/10.1111/ANEC.12111>.
- [8] J.J. Oresko, et al., A wearable smartphone-based platform for real-time cardiovascular disease detection via electrocardiogram processing, *IEEE Trans. Inf. Technol. Biomed.* 14 (3) (May 2010) 734–740, <https://doi.org/10.1109/TTB.2010.2047865>.
- [9] L. Zhao, C. Liu, S. Wei, Q. Shen, F. Zhou, J. Li, A new entropy-based atrial fibrillation detection method for Scanning wearable ECG recordings, *Entropy* 20 (12) (Dec. 2018), <https://doi.org/10.3390/E20120904>.
- [10] H. Li, D. Yuan, Y. Wang, D. Cui, L. Cao, Arrhythmia classification based on multi-domain feature extraction for an ECG recognition system, *Sensors* 16 (10) (Oct. 2016), <https://doi.org/10.3390/S16101744>.
- [11] S. Hong, et al., ENCASE: an ENsemble CLASSifier for ECG classification using expert features and deep neural networks, in: *Computing in Cardiology*, IEEE Computer Society, 2017, pp. 1–4, <https://doi.org/10.22489/CinC.2017.178-245>.
- [12] M. Zabihi, A.B. Rad, A.K. Katsaggelos, S. Kiranyaz, S. Narkilahti, M. Gabbouj, Detection of atrial fibrillation in ECG hand-held devices using a random forest classifier, in: *Computing in Cardiology*, IEEE Computer Society, 2017, pp. 1–4, <https://doi.org/10.22489/CinC.2017.069-336>.
- [13] S. Datta, et al., Identifying normal, AF and other abnormal ECG rhythms using a cascaded binary classifier, in: *Computing in Cardiology*, IEEE Computer Society, 2017, pp. 1–4, <https://doi.org/10.22489/CinC.2017.173-154>.
- [14] R. Mahajan, R. Kamaleswaran, J.A. Howe, O. Akbilgic, Cardiac rhythm classification from a short single lead ECG recording via random forest, in: *Computing in Cardiology*, IEEE Computer Society, 2017, pp. 1–4, <https://doi.org/10.22489/CinC.2017.179-403>.
- [15] M. Kisohara, Y. Masuda, E. Yuda, N. Ueda, J. Hayano, Optimal length of R–R interval segment window for Lorenz plot detection of paroxysmal atrial fibrillation by machine learning, *Biomed. Eng. Online* 19 (1) (Jun. 2020), <https://doi.org/10.1186/S12938-020-00795-Y>.
- [16] C. Huang, S. Ye, H. Chen, D. Li, F. He, Y. Tu, A novel method for detection of the transition between atrial fibrillation and sinus rhythm, *IEEE Trans. Biomed. Eng.* 58 (4) (Apr. 2011) 1113–1119, <https://doi.org/10.1109/TBME.2010.2096506>.

- [17] K. Tatenno, L. Glass, Automatic detection of atrial fibrillation using the coefficient of variation and density histograms of RR and Δ RR intervals, *Med. Biol. Eng. Comput.* 39 (6) (2001) 664–671, <https://doi.org/10.1007/BF02345439/METRICS>.
- [18] M. Henriksson, A. Petrenas, V. Marozas, F. Sandberg, L. Sornmo, Model-based assessment of f-wave signal quality in patients with atrial fibrillation, *IEEE Trans. Biomed. Eng.* 65 (11) (Nov. 2018) 2600–2611, <https://doi.org/10.1109/TBME.2018.2810508>.
- [19] S. Ladavich, B. Ghorraani, Developing an atrial activity-based algorithm for detection of atrial fibrillation, in: 2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC 2014, Nov. 2014, pp. 54–57, <https://doi.org/10.1109/EMBC.2014.6943527>.
- [20] H. Serhal, N. Abdallah, J.-M. Marion, P. Chauvet, M. Oueidat, A. Humeau-Heurtier, Overview on prediction, detection, and classification of atrial fibrillation using wavelets and AI on ECG, *Comput. Biol. Med.* 142 (2022) 105168, <https://doi.org/10.1016/j.combiomed.2021.105168>.
- [21] F. Ma, J. Zhang, W. Liang, J. Xue, Automated classification of atrial fibrillation using Artificial neural network for wearable devices, *Math. Probl Eng.* 2020 (2020), <https://doi.org/10.1155/2020/9159158>.
- [22] S. Somani, et al., Deep learning and the electrocardiogram: review of the current state-of-the-art, *EP Europace* (Feb. 2021), <https://doi.org/10.1093/europace/eaab377>.
- [23] P. Sodmann and M. Vollmer, “ECG Segmentation Using a Neural Network as the Basis for Detection of Cardiac Pathologies”.
- [24] Z. Yue, Z. Jinjing, Atrial fibrillation detection based on EEMD and XGBoost, in: *Journal of Physics: Conference Series*, Institute of Physics Publishing, May 2019, <https://doi.org/10.1088/1742-6596/1229/1/012074>.
- [25] O. Yildirim, M. Talo, E.J. Ciccio, R.S. Tan, U.R. Acharya, Accurate deep neural network model to detect cardiac arrhythmia on more than 10,000 individual subject ECG records, *Comput Methods Programs Biomed* 197 (Dec. 2020), <https://doi.org/10.1016/j.cmpb.2020.105740>.
- [26] G. Sannino, G. de Pietro, A deep learning approach for ECG-based heartbeat classification for arrhythmia detection, *Future Generat. Comput. Syst.* 86 (Sep. 2018) 446–455, <https://doi.org/10.1016/j.future.2018.03.057>.
- [27] A.Y. Hannun, et al., Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network, *Nat Med* 25 (1) (Jan. 2019) 65–69, <https://doi.org/10.1038/s41591-018-0268-3>.
- [28] E. Izci, M.A. Ozdemir, M. Degirmenci, A. Akan, Cardiac arrhythmia detection from 2d ecg images by using deep learning technique, in: *TIPTEKNO 2019 - Tip Teknolojileri Kongresi*, Institute of Electrical and Electronics Engineers Inc., Oct. 2019, <https://doi.org/10.1109/TIPTEKNO.2019.8895011>.
- [29] P. Cao, et al., A novel data augmentation method to enhance deep neural networks for detection of atrial fibrillation, *Biomed. Signal Process Control* 56 (2020) 101675, <https://doi.org/10.1016/j.bspc.2019.101675>.
- [30] M. Hammad, A.M. Iliyasa, A. Subasi, E.S.L. Ho, A.A.A. El-Latif, A Multitier deep learning model for arrhythmia detection, *IEEE Trans. Instrum. Meas.* 70 (2021), <https://doi.org/10.1109/TIM.2020.3033072>.
- [31] M.T. Nguyen, B. van Nguyen, K. Kim, Deep feature learning for Sudden cardiac arrest detection in automated External Defibrillators, *Sci. Rep.* 8 (1) (Dec. 2018), <https://doi.org/10.1038/s41598-018-33424-9>.
- [32] S. Somani, et al., Deep learning and the electrocardiogram: review of the current state-of-the-art, *EP Europace* (Feb. 2021), <https://doi.org/10.1093/europace/eaab377>.
- [33] S. Raghunath, et al., Deep neural networks can Predict new-Onset atrial fibrillation from the 12-lead ECG and help identify those at risk of atrial fibrillation-related stroke, *Circulation* (2021) 1287–1298, <https://doi.org/10.1161/CIRCULATIONAHA.120.047829>.
- [34] U. Satija, B. Ramkumar, M. Sabarimalai Manikandan, A review of signal processing techniques for electrocardiogram signal quality assessment, *IEEE Rev Biomed Eng* 11 (Feb. 2018) 36–52, <https://doi.org/10.1109/RBME.2018.2810957>.
- [35] H. Rajpal, M. Sas, C. Lockwood, R. Joakim, N.S. Peters, M. Falkenberg, Interpretable XGBoost based classification of 12-lead ECGs applying information theory measures from Neuroscience, in: *2020 Computing in Cardiology, Rimini, Italy*, 2020, pp. 1–4, <https://doi.org/10.22489/CinC.2020.185>.
- [36] N. Nuryani, B. Harjito, I. Yahya, and A. Lestari, “Atrial Fibrillation Detection Using Support Vector Machine”, doi: 10.1161/CIR.
- [37] D. Kleyko, E. Osipov, U. Wiklund, A comprehensive study of complexity and performance of automatic detection of atrial fibrillation: classification of Long ECG recordings based on the PhysioNet computing in cardiology challenge 2017, *Biomed Phys Eng Express* 7 (2) (2020), <https://doi.org/10.1088/2057-1976/ab6e1e>.
- [38] K. Weimann, T.O.F. Conrad, Transfer learning for ECG classification, *Sci. Rep.* 11 (1) (Mar. 2021) 5251, <https://doi.org/10.1038/s41598-021-84374-8>.
- [39] T. Golany, D. Freedman, and K. Radinsky, “SimGANs: Simulator-Based Generative Adversarial Networks for ECG Synthesis to Improve Deep ECG Classification”.
- [40] M.U. Zahid, S. Kiranyaz, M. Gabbouj, Global ECG classification by Self-Operational neural networks with feature Injection, *IEEE Trans. Biomed. Eng.* 70 (1) (Jan. 2023) 205–215, <https://doi.org/10.1109/TBME.2022.3187874>.
- [41] J. Xiao, et al., ULECGNet: an Ultra-lightweight end-to-end ECG classification neural network, *IEEE J. Biomed. Health Inform.* 26 (1) (Jan. 2022) 206–217, <https://doi.org/10.1109/JBHI.2021.3090421>.
- [42] A. Rahman, et al., ECG classification for detecting ECG arrhythmia Empowered with deep learning approaches, *Comput. Intell. Neurosci.* 2022 (Jul. 2022) 1–12, <https://doi.org/10.1155/2022/6852845>.
- [43] T. Wang, P. Bonizzi, J. Karel, R. Peeters, ECGI with a deep neural network and 2D normalized Body Surface potential Maps, in: *2021 Computing in Cardiology (CinC)*, IEEE, Brno, Czech Republic, Sep. 2021, pp. 1–4, <https://doi.org/10.23919/CinC53138.2021.9662734>.
- [44] W. Cai, S. Hu, J. Yang, J. Cao, Automatic 12-lead ECG classification using deep neural networks, in: *Presented at the 2020 Computing in Cardiology Conference*, Dec. 2020, <https://doi.org/10.22489/CinC.2020.039>.
- [45] G.D. Clifford, et al., AF classification from a short single lead ECG recording: the PhysioNet/computing in cardiology challenge 2017, in: *Computing in Cardiology, IEEE Computer Society*, 2017, pp. 1–4, <https://doi.org/10.22489/CinC.2017.065469>.
- [46] S. Liaqat, K. Dashtipour, A. Zahid, K. Assaleh, K. Arshad, N. Ramzan, Detection of atrial fibrillation using a machine learning approach, *Information* 11 (12) (Dec. 2020) 1–15, <https://doi.org/10.3390/info11120549>.
- [47] S.H. Jambukia, V.K. Dabhi, H.B. Prajapati, Classification of ECG signals using machine learning techniques: a survey, in: *Conference Proceeding - 2015 International Conference on Advances in Computer Engineering and Applications, ICACEA 2015*, Jul. 2015, pp. 714–721, <https://doi.org/10.1109/ICACEA.2015.7164783>.
- [48] P.S. Addison, Wavelet transforms and the ECG: a review, *Physiol. Meas.* 26 (5) (Oct. 2005), <https://doi.org/10.1088/0967-3334/26/5/R01>.
- [49] R.J. Martis, U.R. Acharya, K.M. Mandana, A.K. Ray, C. Chakraborty, Application of principal component analysis to ECG signals for automated diagnosis of cardiac health, *Expert Syst. Appl.* 39 (14) (Oct. 2012) 11792–11800, <https://doi.org/10.1016/j.eswa.2012.04.072>.
- [50] B.U. Köhler, C. Hennig, R. Orglmeister, The principles of software QRS detection, *IEEE Eng. Med. Biol. Mag.* 21 (1) (2002) 42–57, <https://doi.org/10.1109/51.993193>.
- [51] F. Grizali, G. Frangakis, G. Papakonstantinou, Detection of the P and T waves in an ECG, *Comput. Biomed. Res.* 22 (1) (1989) 83–91, [https://doi.org/10.1016/0010-4809\(89\)90017-7](https://doi.org/10.1016/0010-4809(89)90017-7).
- [52] M. Elgendy, M. Meo, D. Abbott, A Proof-of-Concept study: Simple and effective detection of P and T waves in arrhythmic ECG signals, *Bioengineering* 3 (4) (Dec. 2016), <https://doi.org/10.3390/BIOENGINEERING3040026>.
- [53] G. Berkooz, P. Holmes, J.L. Lumley, The Proper Orthogonal Decomposition in the Analysis of Turbulent Flows, vol. 25, Nov. 2003, pp. 539–575, <https://doi.org/10.1146/ANNUREV.FL.25.010193.002543>, 1.
- [54] L. Sirovich, Turbulence and the DYNAMICS OF COHERENT STRUCTURES part I: COHERENT STRUCTURES, *Q. Appl. Math.* XLV (1987) 561–571.
- [55] N. Ahmed, T. Natarajan, K.R. Rao, Discrete cosine transform, *IEEE Trans. Comput.* C-23 (1) (1974) 90–93, <https://doi.org/10.1109/T-C.1974.223784>.
- [56] E.A. Perez Alday, et al., Classification of 12-lead ECGs: the PhysioNet/Computing in Cardiology Challenge, 2020.
- [57] B. Meyers, M. Brindise, R. Sharmin, J. Zhang, P. Vlachos, Detecting atrial fibrillation in ECGs using correlation-based features and machine learning, *J. Am. Coll. Cardiol.* 79 (9) (2022) 50, [https://doi.org/10.1016/s0735-1097\(22\)01041-5](https://doi.org/10.1016/s0735-1097(22)01041-5).